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RESEARCH ARTICLE

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Glucocorticoids impair T lymphopoiesis after myocardial infarction

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Abstract

The thymus, where T lymphocytes develop and mature, is sensitive to insults such as tissue ischemia or injury. The insults can cause thymic atrophy and compromise T-cell development, potentially impairing adaptive immunity. The objective of this study was to investigate whether myocardial infarction (MI) induces thymic injury to impair T lymphopoiesis and to uncover the underlying mechanisms. When compared with sham controls, MI mice at *day 7* post-MI exhibited smaller thymus, lower cellularity, as well as less thymocytes at different developmental stages, indicative of T-lymphopoiesis impairment following MI. Accordingly, the spleen of MI mice has less T cells and recent thymic emigrants (RTEs), implying that the thymus of MI mice releases fewer mature thymocytes than sham controls. Interestingly, the secretory function of splenic T cells was not affected by MI. Further experiments showed that the reduction of thymocytes in MI mice was due to increased thymocyte apoptosis. Removal of adrenal glands by adrenalectomy (ADX) prevented MI-induced thymic injury and dysfunction, whereas corticosterone supplementation in ADX + MI mice reinduced thymic injury and dysfunction, indicating that glucocorticoids mediate thymic damage triggered by MI. Eosinophils play essential roles in thymic regeneration postirradiation, and eosinophil-deficient mice exhibit impaired thymic recovery after sublethal irradiation. Interestingly, the thymus was fully regenerated in both wild-type and eosinophil-deficient mice at *day 14* post-MI, suggesting that eosinophils are not critical for thymus regeneration post-MI. In conclusion, our study demonstrates that MI-induced glucocorticoids trigger thymocyte apoptosis and impair T lymphopoiesis, resulting in less mature thymocyte release to the spleen.

NEW & NOTEWORTHY The thymus is essential for maintaining whole body T-cell output. Thymic injury can adversely affect T lymphopoiesis and T-cell immune response. This study demonstrates that MI induces thymocyte apoptosis and compromises T lymphopoiesis, resulting in fewer releases of mature thymocytes to the spleen. This process is mediated by glucocorticoids secreted by adrenal glands. Therefore, targeting glucocorticoids represents a novel approach to attenuate post-MI thymic injury.

glucocorticoids; myocardial infarction; T lymphopoiesis; thymocyte; thymus

INTRODUCTION

The thymus is the primary lymphoid organ responsible for generating immunocompetent and self-tolerant T lymphocytes. T-cell development can be divided into intra-bone marrow and intra-thymic stages. In the bone marrow, hematopoietic stem/progenitor cells (HSPCs) develop into multipotent progenitors (MPP), which progress into either common myeloid progenitors (CMP) or common lymphoid progenitors (CLP) (1). CMP will further give rise to granulocyte-monocyte progenitor (GMP) and subsequent myeloid cells (e.g., granulocytes and monocytes), red blood cells, and platelets (2). In contrast, CLP will migrate into the thymus through a CCR7- and CCR9-dependent manner and generate early thymic progenitors (ETPs) (1, 3). ETP will sequentially differentiate into double negative $CD4^-$ $CD8^-$, double positive $CD4^+$ $CD8^+$, and mature single positive $CD4^+$ and $CD8^+$ thymocytes (4). Double negative (DN) thymocytes experience DN1, DN2, DN3, and DN4 phenotypic stages (5). During the developmental stages, thymocytes sequentially undergo positive and negative selection and eventually become major histocompatibility complex (MHC) class-restricted and self-tolerant mature thymocytes (6). Once mature, thymocytes in the thymus are released into the periphery (e.g., blood, spleen, and lymph nodes) as recent thymic emigrants (RTEs) (7). RTEs undergo a period of ~3 wk of postthymic maturation in the periphery before transitioning into mature naïve T cells (8, 9).

The thymus gradually involutes with age, a process known as age-associated thymic involution (10). In humans, thymic involution occurs after the first year of life, with a size reduction at a rate of 3% per year (11). By the age of 55 yr old, only

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5% of naïve T cells originate from thymic export (12). It is currently accepted that the adult thymus contributes to minimal T cell output and has limited functional roles (12). This dogma, however, has recently been challenged. A retrospective clinical study demonstrates that adult patients who underwent thymectomy have a higher all-cause mortality and an increased risk of cancer than matching controls at 5 yr after surgery (13). Moreover, these patients have fewer newly generated T cells. These findings indicate that the adult thymus contributes to new T-cell generation and is vital to maintaining immune competence and overall health. The findings that thymectomy in infants with congenital heart disease is related to long-term impairment of the immune system (14) highlight the importance of the thymus.

The thymus is highly prone to injury, and thymus damage occurs in a wide range of insults, including stroke (15), cardiac arrest (16), infections (17), cancer (18), autoimmune disease (19), graft versus host disease (20), dietary restriction (21), cytoreductive therapy (22), immunosuppressants (23), and steroids (24). These insults can lead to thymic atrophy and impair systemic T-cell output, potentially increasing the risk of infection and autoimmune diseases (25).

Our previous study shows that myocardial infarction (MI) activates the hypothalamic-pituitary-adrenal (HPA) axis to elevate glucocorticoid secretion, and increased glucocorticoids trigger peripheral lymphocytes migrating to the bone marrow, leading to lymphopenia (26). Glucocorticoids can induce thymic damage and impair T-cell lymphopoiesis (24). Accordingly, in this study, we investigated the hypothesis that MI impairs thymocyte development and thymic function by activating the HPA axis and increasing glucocorticoid generation. Our study unravels that MI induces thymocyte apoptosis to impair T lymphopoiesis, leading to less T-cell generation. Removal of adrenal glands protects mice from MI-induced thymic damage, and corticosterone supplementation in adrenalectomized MI mice reinduces thymic injury, indicating that glucocorticoids impair T lymphopoiesis after MI.

METHODS

Mice

C57BL/6J wild-type (WT, Strain No. 000664) and ΔdblGATA mice deficient for eosinophils (B6.129S1(C)-Gata1tm6Sho/LvtzJ., Strain No. 033551) were purchased from the Jackson Laboratory and bred in animal facility at the University of South Florida. Adult (4–8 mo old) male and female mice were used in this study. Animals were housed in a facility with a 12-h:12-h light/dark cycle and had free access to water and standard rodent chow. Mice that had undergone adrenalectomy were provided with saline. All animal procedures were approved by the Institutional Animal Care and Use Committee at the University of South Florida and were conducted in accordance with the *Guide for the Care and Use of Laboratory Animals*.

Myocardial Infarction

MI was induced by permanent occlusion of the left anterior descending (LAD) coronary artery, as described previously (26, 27). Sustained release buprenorphine (1–1.5 mg/kg) was administered subcutaneously before surgery. Mice were anesthetized with 1–3% inhalational isoflurane, intubated, and ventilated with a standard rodent ventilator (Harvard Apparatus). The thoracic cavity was opened through an incision between the third and fourth ribs. The LAD coronary artery was ligated with 8-0 nylon suture (AD Surgical, Cat. No. XXS-N807T6) placed 1–2 mm distal to the left atrium. Infarction was confirmed by visualizing myocardium blanching. The thoracic cavity and skin were closed using a 6-0 polypropylene suture (AD Surgical, Cat. No. XS-P618R11). Sham mice underwent the same surgical procedure without artery ligation.

Euthanasia

During euthanasia procedure, mouse blood was collected into EDTA tubes (BD Biosciences, 365974) from the right ventricle via cardiac puncture. The mice were then flushed with cold PBS to remove residual blood, and the thymus, spleen, bone marrow, and heart were harvested.

Flow Cytometry

Mouse thymocytes were isolated by mashing thymic tissues between the frosted ends of two microscope slides. Single-cell suspension was labeled with antibodies against CD45 (BioLegend, 103128), CD11b (BioLegend, 101243), CD11c (BioLegend,117349), CD4 (BioLegend, 100478), CD8 (BioLegend,155020), CD25 (BioLegend, 102043), and CD44 (BioLegend, 103008). For ETP, thymocytes were stained with antibodies against the mouse lineage cocktail (BioLegend, 133303), c-kit (BioLegend, 105813), CD25, IL-7Rα (BioLegend, 158212), and CD44 (BioLegend, 103012).

The splenic cell suspension was made by mashing tissues between the frosted ends of two microscope slides. The erythrocytes were removed by adding 0.2% hypotonic NaCl solution, followed by the addition of an equal volume of 1.6% NaCl solution (26). The single-cell suspension was labeled with antibodies against CD45, CD11b, CD3 (BioLegend, 100228), CD4, and CD8. For splenic RTEs, splenocytes were stained with antibodies against CD45, CD11b, CD3, CD4, CD8, and Qa2 (BioLegend, 121714).

Bone marrow cells were collected from the mouse femur by centrifugation at 10,000 *g* for 30 s, and red blood cells were lysed with red blood cell lysis buffer (BioLegend, 420301) (26). The single cell suspension was stained with antibodies against mouse lineage cocktail, Sca-1 (BioLegend, 108111), c-kit, Flt3 (BioLegend, 135311), IL-7Rα, CD34 (BioLegend, 152208), and CD16/32 (BioLegend, 101337).

Whole blood was stained with a panel of antibodies against CD45, CD11b, CD3, CD4, and CD8 for 20 min at room temperature, followed by incubation with red blood cell lysis/fixation solution (BioLegend, 422401) for 15 min at room temperature. For blood RTE evaluation, blood was stained with antibodies against CD45, CD11b, CD3, CD4, CD8, and Qa2.

For all flow cytometric experiments except for blood, dead cells were excluded using Zombie NIR fixable viability dye (BioLegend, 423105). The samples were run on a Cytek Northern Lights flow cytometer (CYTEK), and data were analyzed using SpectroFlo and FlowJo software. The total cell count in the tissue was quantified using a Luna-II automated cell counter (Logos Biosystems), and the numbers of different cells were calculated based on relative abundance obtained by flow cytometry (number = total cell count × percentage of positive cells). Supplemental Table S1 lists markers used to identify different cells. All supplemental material is available at https://doi.org/10.6084/m9.figshare.25499596.v4.

T-Cell Isolation and Stimulation

On *day 7* after MI or sham surgery, splenic T cells were isolated using a pan T-cell isolation kit II (Miltenyi Biotec, 130-095-130). Briefly, single splenocytes were incubated with a cocktail of biotin-conjugated antibodies against CD11b, CD11c, CD19, CD45R (B220), CD49b (DX5), CD105, anti-MHC-class II, and Ter-119, followed by incubation with anti-biotin microbeads. The cell suspension was applied over a magnetic LS column (Miltenyi Biotec, 130-042-401). Non-T cells were retained in the magnetic column, while T cells flew through the column and were collected. T cells were then stimulated with phorbol 12-myristate 13-acetate (PMA, Cayman, 10008014, 20 ng/mL) plus ionomycin (Cayman, 11932, 0.5 μg/mL) or vehicle for 4 h, as described by others (16).

Real-Time PCR

Semiquantitative real-time PCR was carried out as previously described (28). Briefly, total RNA was extracted from splenic T cells using TRIzol (Thermo Fisher Scientific, 15596026) and PureLink RNA Kit (Thermo Fisher Scientific, 12183018 A). RNA concentration was measured using NanoDrop One (Thermo Fisher Scientific). Reverse transcription of RNA was run on a PTC Tempo 48/48 Thermal Cycler (Bio-Rad) using a high-capacity RNA-to-cDNA Kit (Thermo Fisher Scientific, 4387406). Gene expression was measured using Taqman gene expression master mix (Thermo Fisher Scientific, 4369016) plus individual primers for *Ifnγ*, *Tnfα*, *Il2*, *Il4*, *Il17*, and *granzyme b*. The PCR mix was run on a Bio-Rad CFX Connect real-time system. The gene levels were normalized to the reference gene *hprt1* and expressed as fold change to control groups. MIQE guidelines were followed for all the PCR experiments and analysis (29).

Evaluation of Cell Proliferation

Single-cell suspension of thymocytes was prepared as described above. To evaluate thymocyte proliferation, 0.5 μL of BrdU (10 mg/mL) was added to 1 mL of thymic cell suspension and cultured for 1 h at 37°C with 5% CO_2. For thymocyte evaluation, cells were stained with antibodies against CD45, CD11b, CD11c, CD4, CD8, CD25, and CD44. For ETP assessment, cells were stained with a mouse lineage cocktail, c-kit, CD25, IL-7Rα, and CD44. Cells were then fixed and permeabilized for intracellular staining of BrdU using the phase-flow kit (BioLegend, 370706) according to the manufacturer's instructions. Proliferation was calculated as the percentage of BrdU$^+$ cells to total cells.

Evaluation of Cell Apoptosis

Thymocytes and ETP were labeled as described above. Cell apoptosis was then evaluated using the annexin V apoptosis detection kit (BioLegend, 640930). Apoptosis was calculated as the percentage of annexin V$^+$ cells to total cells.

Adrenalectomy and Corticosterone Feeding

Mice were anesthetized with 1–3% isoflurane. Sustained release buprenorphine (1–1.5 mg/kg) was administered subcutaneously before surgery. A flank incision just below the ribs on both sides was made. Two adrenal glands were exposed, dissected from the surrounding fat pad, and removed (26, 30). The abdominal cavity and skin were closed using a 6-0 polypropylene suture. After surgery, all animals were provided with saline in drinking bottles to maintain sodium balance. Sham mice underwent the same procedure without removing adrenal glands.

To feed mice, corticosterone (Sigma-Aldrich, C2505, 70 μg/mL, equivalent to 10 mg/kg/day) was dissolved in saline with 0.45% 2-hydroxypropyl-β-cyclodextrin (Sigma-Aldrich, 332593) and provided in lightproof drinking bottles (26, 31). Corticosterone water was prepared fresh every 3 days to prevent degradation.

Statistical Analysis

Data are expressed as means ± SE. For data that followed Gaussian distribution, multiple group comparisons were performed using ordinary one-way ANOVA and Tukey's multiple comparisons test when SD was equal among groups or using Brown–Forsythe and Welch ANOVA and Dunnett's T3 multiple comparisons test when SD was not equal among groups. Two group comparisons were analyzed with a parametric unpaired *t* test when both groups had the same SD or a parametric unpaired *t* test with Welch's correction when the SD was not equal in both groups. For data that did not follow Gaussian distribution, multiple group comparisons were analyzed with the nonparametric Kruskal–Wallis test and Dunn's multiple comparisons test, and two group comparisons will be analyzed with the nonparametric test and Mann–Whitney test or Kolmogorov–Smirnov test. A value of $P < 0.05$ was considered statistically significant. Statistical analyses were performed with GraphPad Prism 10 software.

RESULTS

MI Impairs T Lymphopoiesis

The thymus, where T cells develop and mature, is essential for establishing and maintaining whole body T-cell output. To investigate whether MI causes thymic injury that may impair T-cell development, we compared thymic mass, thymic index (thymus weight/body weight), as well as cellularity between sham and *day 7* MI mice. *Day 7* after MI was selected based on the fact that T lymphocytes are relatively late players and their infiltration into the ischemic heart peaks at *day 7* (32). As shown in Fig. 1*A*, the *day 7* MI mice demonstrated a markedly smaller thymus, reduced thymic index, and less cellularity than sham controls, indicating that MI induces thymic injury. To determine which types of thymocytes are involved in MI-induced cell loss, we quantified all developmental thymocytes, including CD4$^-$ CD8$^-$ (DN1, DN2, DN3, and DN4), CD4$^+$ CD8$^+$, CD4$^+$, and CD8$^+$

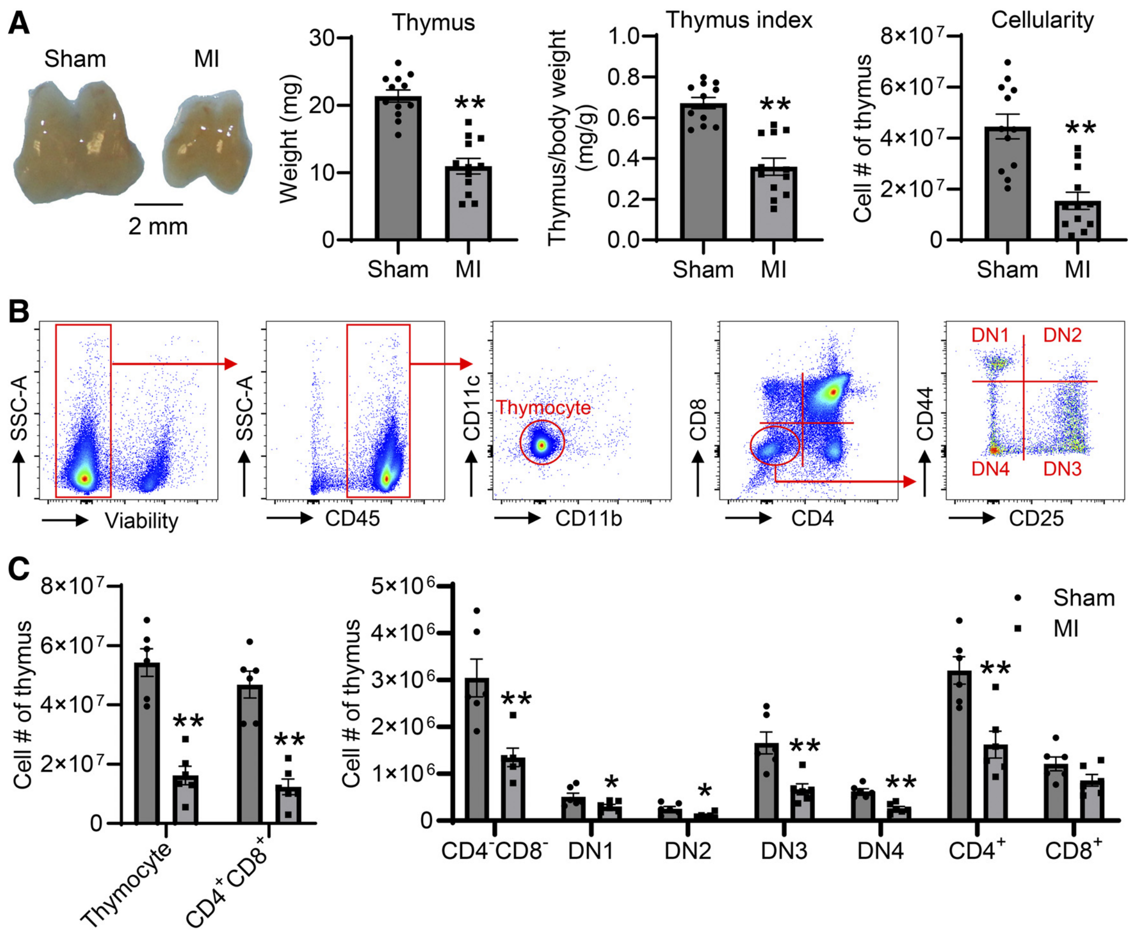

Figure 1. Myocardial infarction (MI) impairs T lymphopoiesis. *A*: *day 7* MI mice exhibit smaller thymus, lower thymus index, and fewer cell count than sham controls; $n = 12$/group. $^{**}P < 0.01$ vs. sham. Unpaired *t* test was used. *B*: gating strategy to identify distinct developing thymocytes with multiparameter flow cytometry. DN, double negative. *C*: all developmental thymocytes, except $CD8^+$ subtypes, are significantly reduced at *day 7* post-MI; $n = 6$/group. $^{*}P < 0.05$, $^{**}P < 0.01$ vs. sham. Unpaired *t* test was used.

thymocytes using multiparameter flow cytometry (Fig. 1*B*). All developing thymocyte subsets, except $CD8^+$ subtypes, were significantly reduced after MI (Fig. 1*C*). This indicates that MI impairs T-cell development from premature double negative until mature single positive stage.

MI Causes a Reduced T-Cell Reservoir in the Spleen

The spleen is the largest secondary lymphoid organ in the body, where abundant T cells are stored (33). Since MI induces thymic injury, we wanted to know whether MI causes less T-cell storage in the spleen. We, therefore, quantified T cells in the spleen of *day 7* MI and sham mice. Although MI mice displayed splenomegaly and higher spleen index, cellularity between sham and MI mice did not differ (Fig. 2*A*). More importantly, the spleen of MI mice contained less T lymphocytes than sham controls (Fig. 2*B*). Interestingly, this is mainly due to the loss of $CD4^+$ but not $CD8^+$ T cells.

RTEs are peripheral T cells newly released from the thymus and can serve as an indicator of thymic function (34). Qa2, a member of nonclassical MHC class Ib molecules, plays a role in antigen presentation (35). The expression level of Qa2 is lower in RTEs than in mature T cells (34). Our data showed that splenic RTE numbers were lower in MI mice than in sham controls (Fig. 2*C*). Taken together, these data imply that MI induces thymic dysfunction to release less mature thymocytes to the periphery, causing less RTEs in the spleen.

We previously showed that circulating T-cell numbers are downregulated at *days 1*, *3*, and *5* post-MI (26). Interestingly, at *day 7* post-MI, there was no significant difference in total and RTE T-cell count between sham and MI mice (Supplemental Fig. S1 *A*–*C*). These data indicate that blood T-cell numbers return to baseline levels by 7 days post-MI.

MI Does Not Affect Secretory Function of Splenic T Cells

Cardiac arrest and resuscitation induce not only thymic output reduction but also functional impairment of peripheral T cells, evidenced by less production of interferon (IFN)-γ and tumor necrosis factor (TNF)-α after stimulation (16). For this reason, we evaluated whether MI also causes T-cell functional defects. We isolated splenic T cells from *day 7* sham and MI mice and measured the expression of T cell-related cytokines after stimulation with PMA plus ionomycin or vehicle. Figure 3*A* showed that the purity of isolated splenic T cells was ∼99%. PMA plus ionomycin treatment

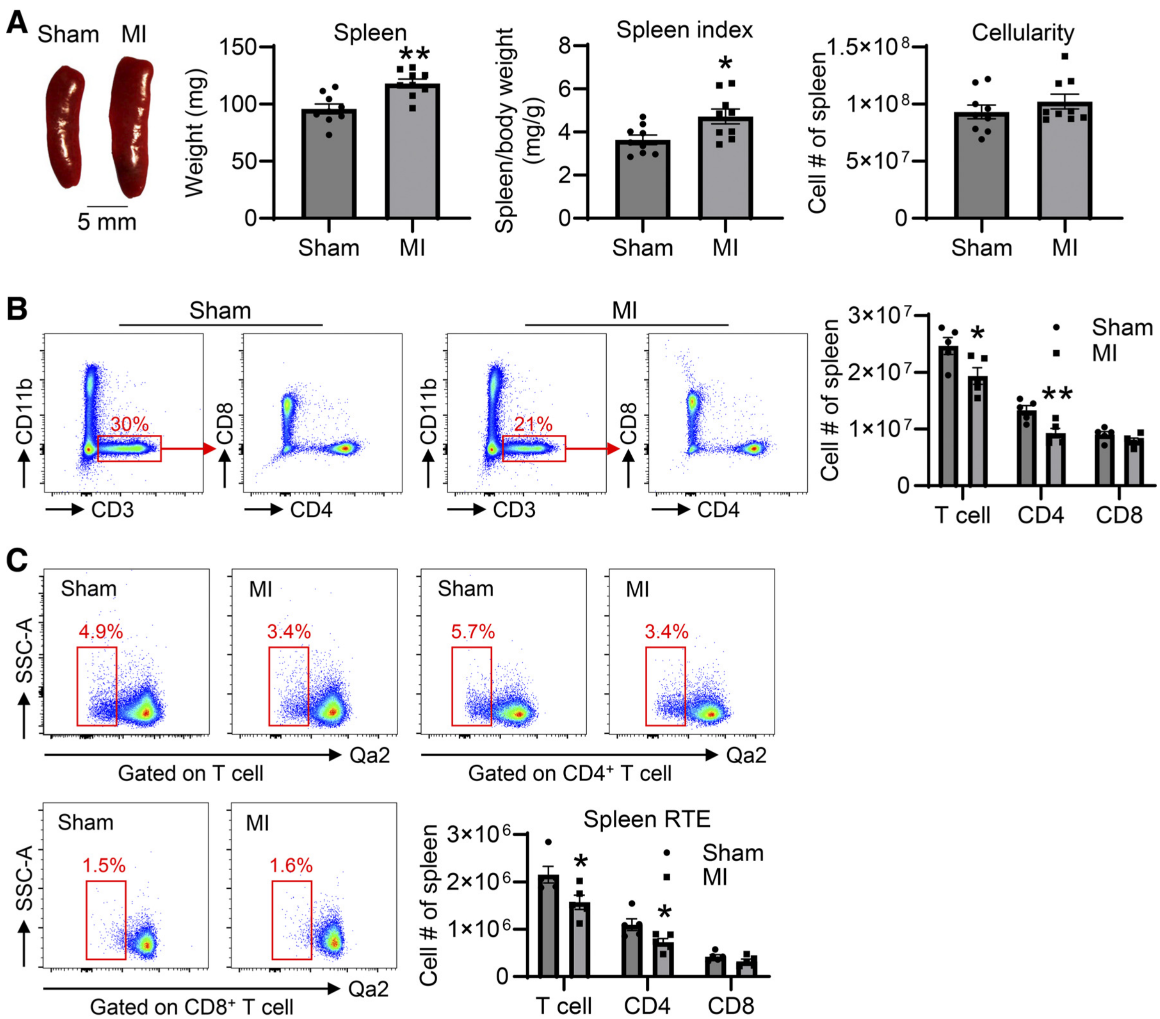

Figure 2. Myocardial infarction (MI) causes a reduction in splenic T-cell pool. *A*: MI mice show splenomegaly and higher spleen index than sham controls; $n = 9$/group. *$P < 0.05$, **$P < 0.01$ vs. sham. Unpaired *t* test was used. *B*: the spleen of *day 7* MI mice shows less total and $CD4^+$ T cells than that of sham mice; $n = 5$/group. *$P < 0.05$, **$P < 0.01$ vs. sham. Unpaired *t*-test was used. *C*: *day 7* MI mice display fewer total and $CD4^+$ RTE T cells in the spleen; $n = 5$/group. *$P < 0.05$ vs. sham. Unpaired *t* test was used.

dramatically increased the expression of *Ifnγ*, *Tnfα*, *Il2*, *Il4*, *Il17*, and *granzyme b* in both sham and MI groups (Fig. 3*B*). Splenic T cells isolated from sham and MI mice demonstrated no difference in response to PMA plus ionomycin stimuli (Fig. 3*B*). This indicates that MI does not compromise T-cell secretory function.

MI Does Not Affect T-Cell Lineage in the Bone Marrow But Leads to Lower ETP Numbers

It is shown that stroke causes B-lymphopoiesis arrest at the pre-B stage in the bone marrow (36). To dissect whether the thymic injury in MI mice is derived from the upstream developmental defect, we evaluated their bone marrow progenitors, including CLP, CMP, and GMP. Interestingly, there was no significant difference in the count of CLP, CMP, and GMP between *day 7* MI and sham mice (Fig. 4*A*), indicative of no developmental defect in the bone marrow of MI mice. The ETP number is, however, significantly less in MI mice than sham controls (Fig. 4*B*), suggesting that the impairment of T-cell development mainly occurs in the thymus, not in the bone marrow.

ETP and Thymocyte Proliferation Are Not Affected by MI

ETP are the progenitor cells that will differentiate into thymocytes. We have so far shown that *day 7* MI mice had less ETP and thymocyte count. To determine whether their reduction was due to decreased cell proliferation, we evaluated their proliferation rate using a BrdU flow cytometry assay. Interestingly, both ETP and thymocytes displayed a similar proliferative rate between MI and sham mice (Fig. 5, *A* and *B*), suggesting that cell proliferation did not contribute to thymic injury induced by MI.

MI Increases Thymocyte Apoptosis

Thymocyte apoptosis contributes to thymic atrophy in sepsis (4) and *Angiostrongylus cantonensis* infection (17). Accordingly, we evaluated ETP and thymocyte apoptosis using flow cytometry with annexin V as an apoptotic marker. ETP apoptosis was comparable between MI and sham mice (Fig. 5*C*); however, MI significantly increased thymocyte apoptosis (Fig. 5*D*). Further analysis revealed that all developmental thymocytes, except for DN1, showed increased

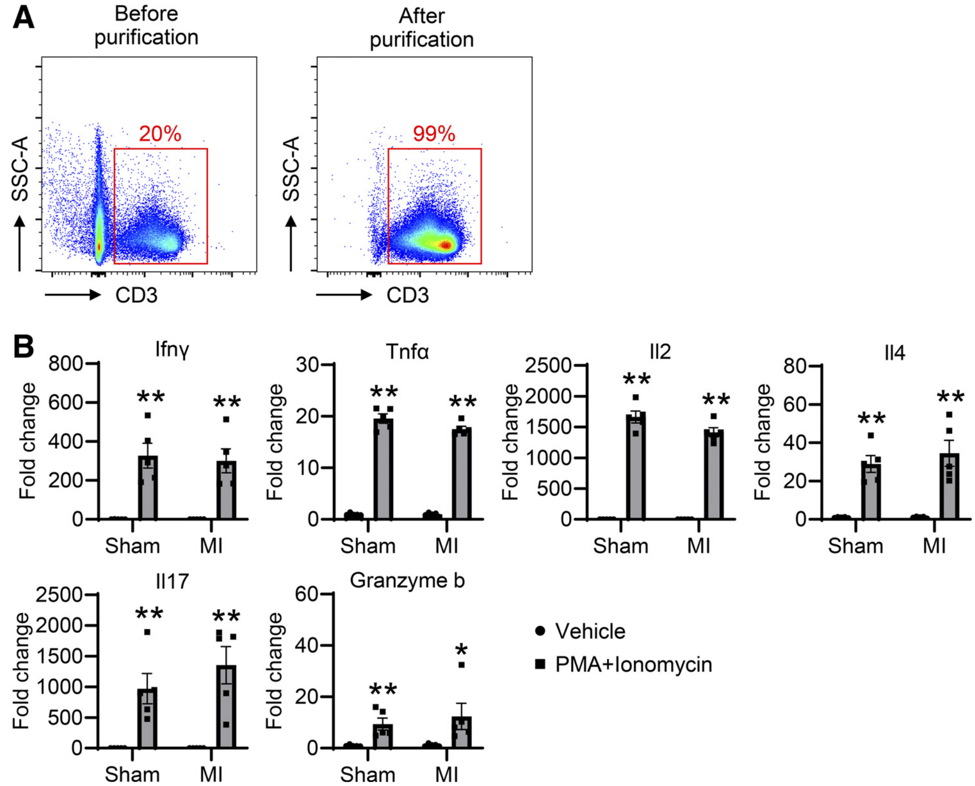

Figure 3. Myocardial infarction (MI) does not affect T-cell secretory function. *A*: purity of isolated splenic T cells was ~99%. *B*: phorbol 12-myristate 13-acetate (PMA) plus ionomycin treatment increased the expression of cytokines in sham and MI T cells to similar levels; $n = 5$/group. $*P < 0.05$, $**P < 0.01$ vs. corresponding vehicle controls. One-way ANOVA was used.

apoptosis after MI (Fig. 5*E*). Consistent with the findings shown in Fig. 1*C*, thymic atrophy is due to the loss of most developmental thymocytes.

Glucocorticoids Mediate Thymic Injury and Dysfunction after MI

It is well known that glucocorticoids can induce thymocyte apoptosis and thymic atrophy. Injection of dexamethasone, a form of synthetic glucocorticoids, causes loss of thymocytes, which is mainly due to the apoptosis of double-positive CD4$^+$ CD8$^+$ thymocytes (24). Infection with *Trypanosoma cruzi* (37) or cardiac arrest injury (16) can activate the HPA axis to promote glucocorticoid secretion and induce thymic atrophy. Our previous study showed that the HPA axis is activated, and plasma glucocorticoid levels are significantly increased after MI (26). Therefore, we hypothesized that glucocorticoids mediate thymic injury induced by MI. To address this hypothesis, we performed adrenalectomy (ADX) on mice to remove two adrenal glands or sham surgery. After 2–4 days, ADX mice underwent MI or sham surgery.

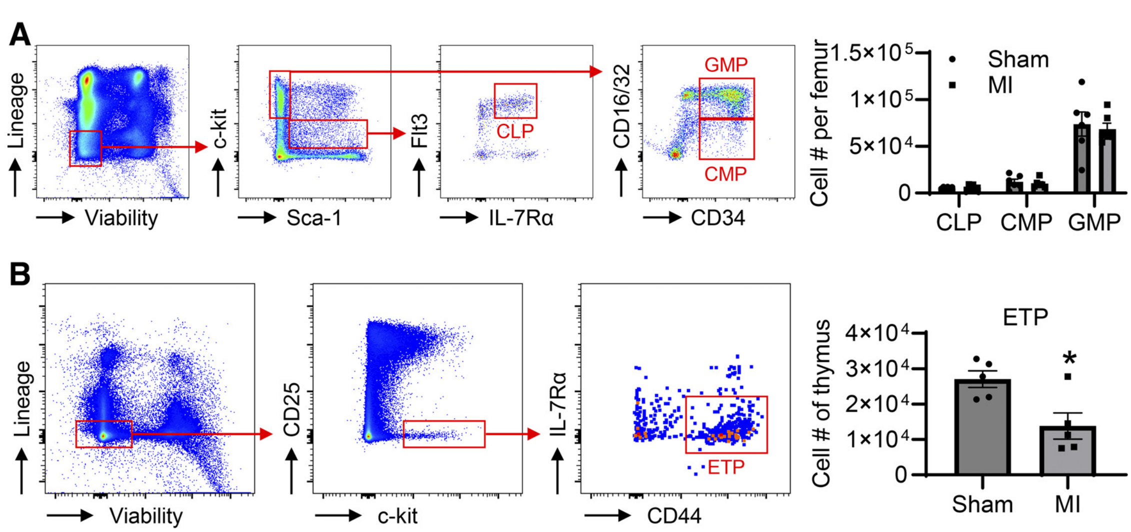

Figure 4. Myocardial infarction (MI) does not affect T-cell lineage in the bone marrow but leads to lower early thymic progenitor (ETP) numbers. *A*: sham and *day 7* MI mice demonstrate similar numbers of common lymphoid progenitors (CLP), common myeloid progenitors (CMP), and granulocyte-monocyte progenitors (GMP) in the bone marrow; $n = 6$/group. Unpaired *t* test was used. *B*: *day 7* MI mice show fewer ETPs in the thymus than sham controls; $n = 5$/group. $*P < 0.05$ vs. sham. Unpaired *t*-test was used.

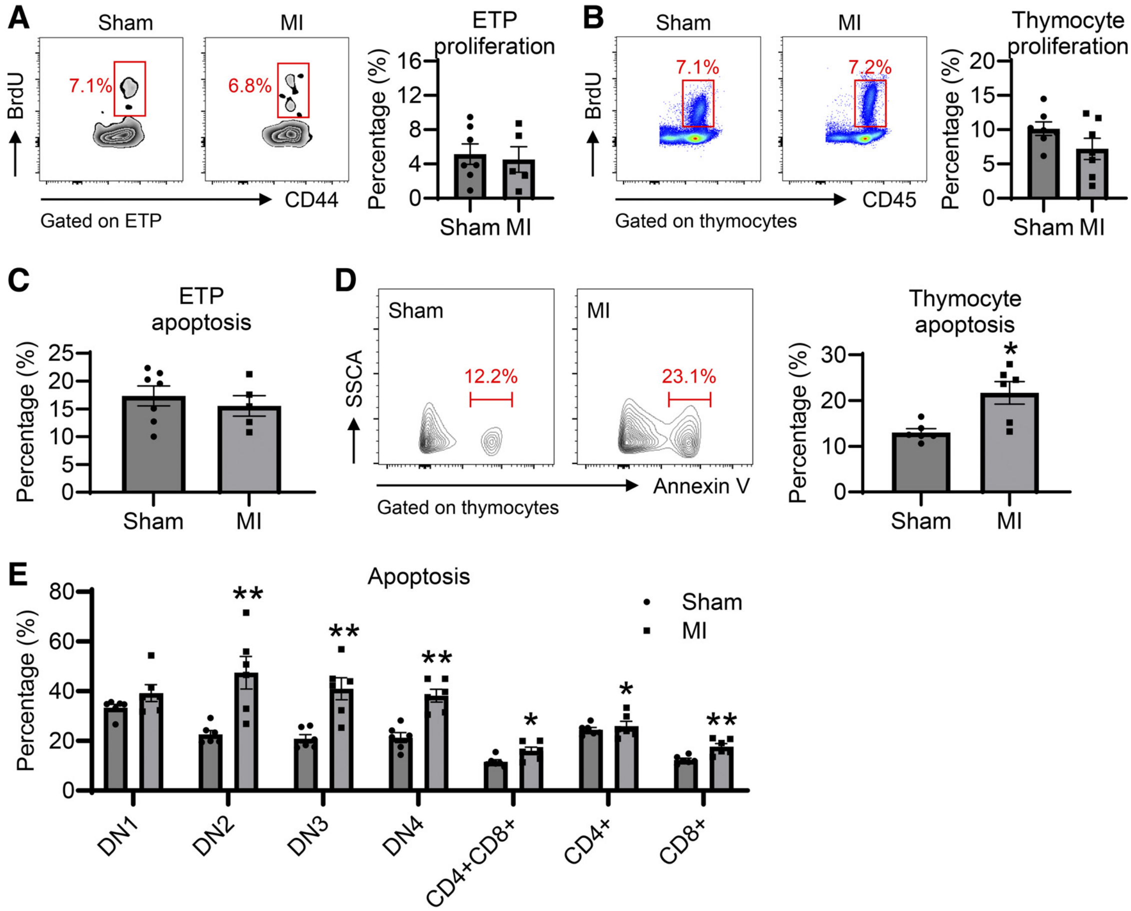

Figure 5. Myocardial infarction (MI) increases thymocyte apoptosis, without affecting the proliferation of early thymic progenitors (ETPs) and thymocytes. *A*: ETP proliferation is comparable between sham and *day 7* MI mice; $n = 5–7$/group. *B*: thymocyte proliferation is similar between sham and *day 7* MI mice; $n = 7$/group. *C*: MI does not affect ETP apoptosis; $n = 5–7$/group. *D*: MI significantly increases thymocyte apoptosis. *E*: MI increases the apoptotic rate of all developing thymocytes, except for double negative 1 (DN1); $n = 6$/group. *$P < 0.05$, **$P < 0.01$ vs. sham. Unpaired *t*-test was used.

Seven days after MI, we compared thymic weight, index, cell count, and splenic total and RTE T cells.

Consistent with published findings (38), ADX per se increased thymic weight, cellularity, and splenic T-cell pool (Fig. 6, *A–D*). MI did not induce thymic injury and impair the splenic T-cell pool when the adrenal glands were removed (Fig. 6, *A–D*), implying that hormones secreted by adrenal glands may mediate MI-induced T-lymphopoiesis impairment. When compared with MI mice, ADX + MI mice displayed a significant increase in thymic weight, thymus index, cellularity, and all developing thymocyte subtypes (Fig. 6, *A* and *B*). Moreover, ADX + MI mice demonstrated more total and RTE T cells in the spleen than MI mice (Fig. 6, *C* and *D*). In contrast, ADX sham surgery had minimal effects on thymic weight, cellularity, and splenic T-cell pool after MI.

Adrenal glands secrete multiple hormones, such as glucocorticoids, aldosterone, androgen, epinephrine, and norepinephrine. To determine the specific contribution of glucocorticoids to MI-induced thymic dysfunction, ADX + MI mice were supplemented with corticosterone in drinking water. Corticosterone supplementation to ADX + MI mice reinduced thymic injury and reduction of splenic T-cell pool in ADX + MI mice (Fig. 6, *A–D*), phenocopying thymic injury induced by MI. These findings confirm the relative importance of glucocorticoids in mediating thymocyte apoptosis, thus impairing T lymphopoiesis in the thymus and decreasing the T-cell pool in the spleen.

Eosinophils Are Dispensable for Thymic Regeneration Following MI

Endogenous thymic regeneration is essential for the renewal of immune competence after injury (39). The thymus is not only sensitive to damage but also has a remarkable capacity for self-regeneration. The thymus can be fully regenerated after injury in immunocompetent mammals (22, 39). Our above data revealed that *day 7* MI mice show thymic injury and dysfunction. To determine whether the thymus can be totally recovered after MI, we investigated *day 14* post-MI. As shown in Fig. 7*A*, *day 7* MI mice demonstrated thymic atrophy; however, the thymus weight, index, and cellularity completely recovered to baseline levels at *day 14* after MI.

Recent studies have shown that eosinophils are essential in tissue repair after injury. Eosinophil deficiency exacerbates aberrant cardiac remodeling and dysfunction post-MI, which could be partially rescued by recombinant IL-4 and cationic protein mEar1 (40, 41). Eosinophils also play indispensable roles in thymic regeneration after sublethal irradiation (42).

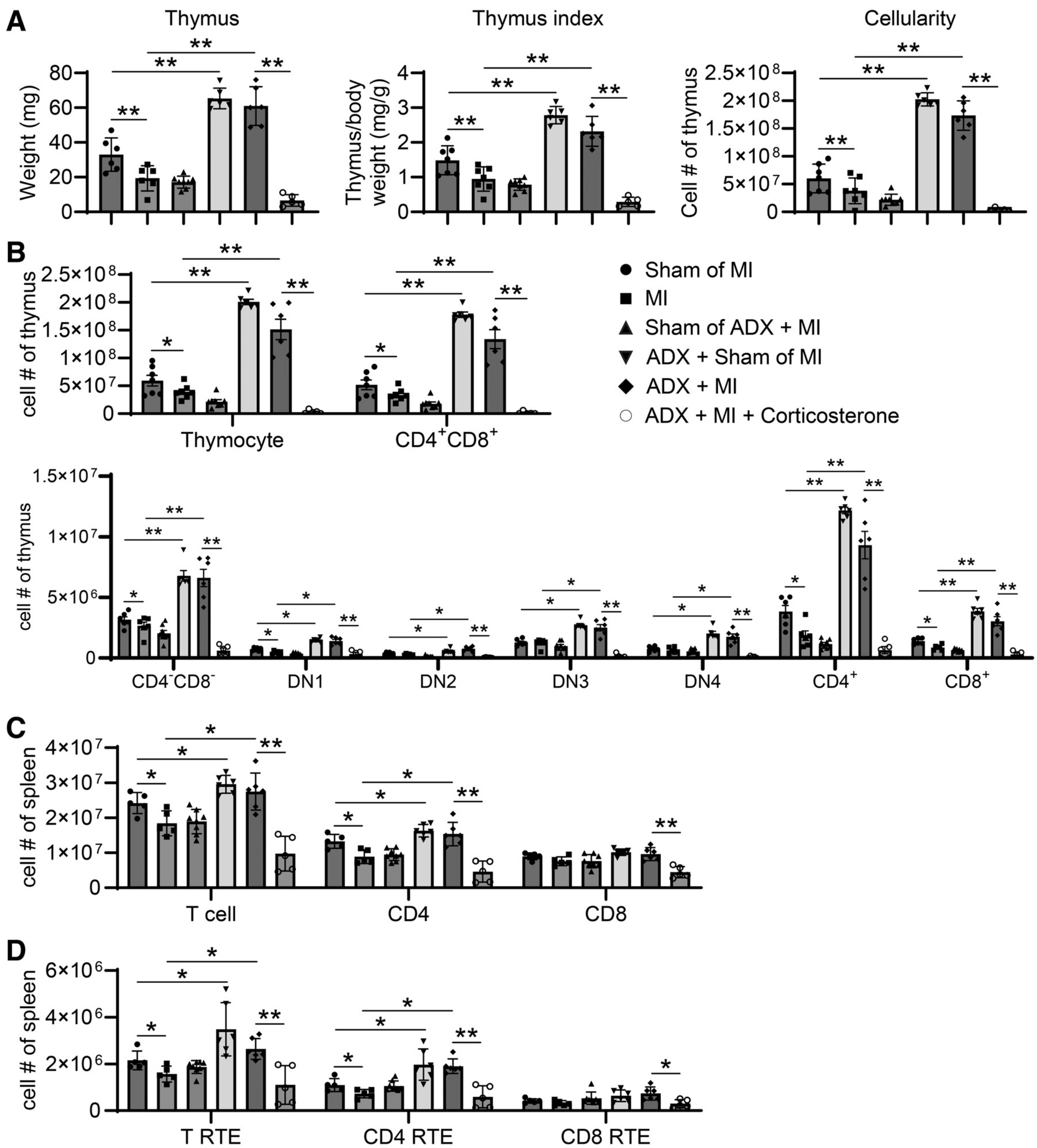

Figure 6. Glucocorticoids mediate myocardial infarction (MI)-induced thymic injury and dysfunction. *A* and *B*: adrenalectomy (ADX) prevents MI-induced thymic injury, and corticosterone supplementation in ADX + MI mice reinduces thymic injury. *C* and *D*: ADX prevents MI-induced reduction in splenic T-cell pool, and corticosterone supplementation in ADX + MI mice reinduces a decrease in splenic T-cell pool; $n = 5–8$/group. $*P < 0.05$, $**P < 0.01$. One-way ANOVA was used.

Wild-type (WT) mice show full recovery of the thymus 35 days after sublethal irradiation; however, eosinophil-deficient mice have significantly lower thymus weight and less count of thymocytes and epithelial cells (42). We next wanted to know whether eosinophils play a role in thymic regeneration post-MI. We evaluated *day 14* post-MI, the time point when WT mice displayed full thymic regeneration. Under unstressed conditions, thymus weight, cellularity, thymocyte count, and blood T cells were comparable between WT and ΔdblGATA eosinophil-deficient mice (Supplemental Fig. S2, *A*–*C*). ΔdblGATA mice had dramatically lower circulating eosinophils than WT mice, confirming the eosinophil deficiency phenotype (Supplemental Fig. S2*C*). Unexpectedly, at *day 14* post-MI, there was no significant difference in thymus weight, thymic index, cellularity, and the count of distinct developmental thymocytes between WT and ΔdblGATA mice (Fig. 7, *B* and *C*). These findings indicate that eosinophils are not required for post-MI thymic regeneration.

DISCUSSION

A wide range of insults or stress can induce thymic injury. Thymic injury impairs the output of thymocytes and adversely affects the adaptive T-cell immune response. Here, we showed

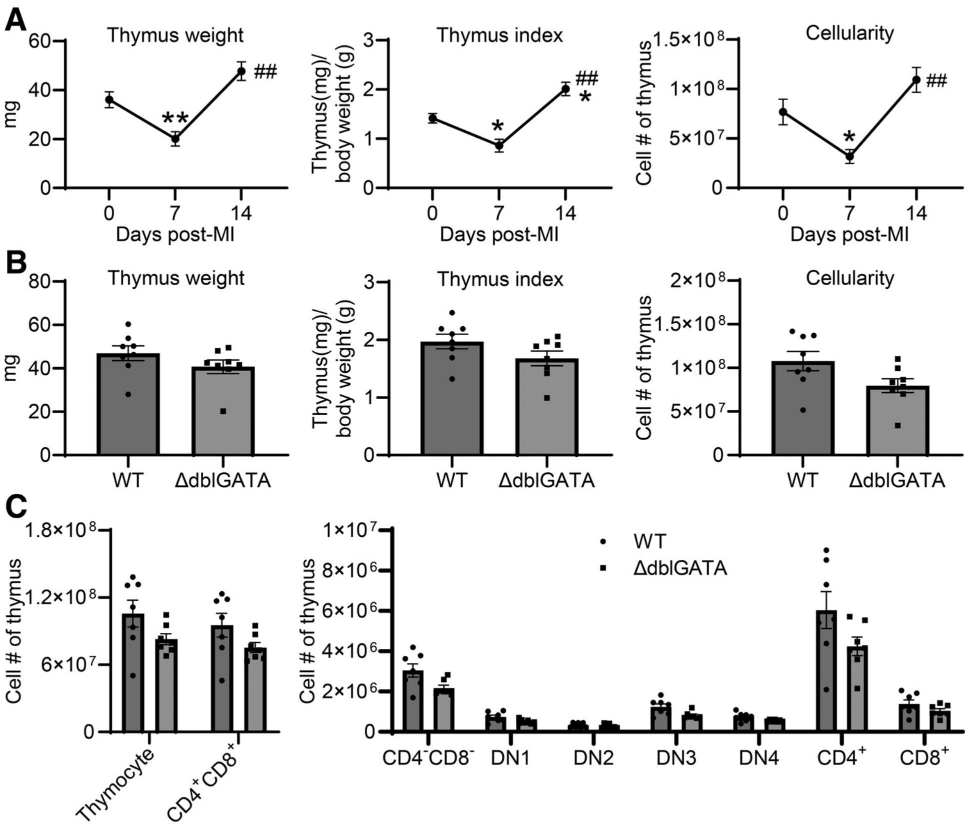

Figure 7. Eosinophils are dispensable for thymic regeneration post-myocardial infarction (MI). *A*: MI induces thymic damage at *day 7* post-MI, which was totally recovered by *day 14* after MI; $n = 7–8$/group. *$P < 0.05$, **$P < 0.01$ vs. *day 0*; ##$P < 0.01$ vs. *day 7*. One-way ANOVA was used. *B*: no significant difference in thymus weight, thymus index, and cellularity between wild-type (WT) and ΔdblGATA eosinophil-deficient mice at *day 14* post-MI; $n = 8$/group. *C*: no significant difference in the count of developing thymocytes between WT and ΔdblGATA mice at *day 14* post-MI; $n = 7$/group. Unpaired *t*-test was used.

that *1*) MI impairs T lymphopoiesis, causing fewer T-cell accumulation in the spleen; *2*) defects in T-cell development after MI mainly occur in the thymus, not in the upstream bone marrow; and *3*) regarding the underlying mechanisms, MI activates the HPA axis and elevates glucocorticoid secretion, and glucocorticoids trigger thymocyte apoptosis and impair T-cell development. Our study, to our knowledge, provides the first line of evidence for MI-induced impairment in T-cell development and decrease in T-cell thymic output.

At a steady state, HSPCs have a balanced differentiation toward myeloid or lymphoid lineage. During the acute inflammatory phase (within 3 days) post-MI, HSPCs display a bias toward myeloid lineage differentiation that generates abundant neutrophils and monocytes, referred to as reactive or emergency hematopoiesis (43). This is, however, usually at the expense of diminished generation of lymphocytes. Our data demonstrated that by *day 7* post-MI, sham and MI mice showed no difference in the numbers of CMP, GMP, and CLP. In contrast, the ETP number in the thymus is lower in MI mice than in sham controls. Interestingly, ETP proliferation and apoptosis were not affected by MI. This implies that some unknown mechanisms contribute to lower ETP post-MI. A previous study shows that ETP in septic mice displays impaired chemotactic capacity, which results in a decrease in the percentage and absolute number of ETP (4). It is likely that ETP in the MI setting may also have chemotactic impairment, which needs to be investigated in the future.

To date, synthetic glucocorticoids remain one of the most powerful anti-inflammatory drugs, used in the treatment of excessive inflammation, autoimmune disease, and organ transplantation rejection. Dexamethasone has been shown to induce thymocyte apoptosis to impair thymopoiesis (24). Our study showed that MI-induced thymic injury is mainly triggered by the HPA axis activation and glucocorticoid generation that induces thymocyte apoptosis. In contrast, radiation-induced thymic damage is partly due to decreased secretion of interleukin (IL)-6, IL-7, and IL-15, which indirectly affect thymocyte survival (25). The different mechanisms of thymic injury induced by radiation and MI may, at least partially, explain the discrepancy in the role of eosinophils in thymic regeneration. In radiation-induced thymic injury, eosinophils cooperate with natural killer T (NKT) cells and type-2 innate lymphoid cells (ILC2) to regulate thymic regeneration (42). After irradiation, thymic endothelial cells upregulate bone morphogenetic protein 4 (BMP4) to promote thymic epithelial recovery and downstream thymic regeneration (22). Tuft cells and fibroblasts promote thymus recovery via ILC2-mediated type-2 immune response in glucocorticoid-induced thymic injury model (44). It would be interesting to know whether tuft cells, fibroblasts, and endothelial cells are involved in thymic regeneration following MI.

We have previously shown that MI does not increase blood lymphocyte apoptosis (26), arguing against apoptosis as a contributor to MI-induced lymphopenia. Instead, blood lymphocytes migrate to the bone marrow in response to MI, causing lymphopenia (26). The current study reveals that MI also impairs T-cell development and thymic output, causing a smaller T-cell pool in the spleen, particularly fewer $CD4^+$ T cells. At *day 7* post-MI, abundant $CD4^+$ T cells infiltrate the ischemic heart (32), which may also consume the T-cell pool in the spleen. However, blood CD4$^+$ T-cell count returns to normal levels by *day 7* post-MI. The spleen enlargement in *day 7* MI mice might be associated with enhanced extramedullary hematopoiesis, increased accumulation of neutrophils and monocytes, and splenic structural alterations (33, 45, 46). T lymphocytes, especially CD4$^+$ T cells, contribute to cardiac wound healing response post-MI (47). Therefore, it is likely that lymphopenia may adversely affect cardiac repair because of T-cell deficiency. This concept is indirectly supported by clinical studies showing that patients with lymphopenia show worse outcome and prognosis (48, 49). Thus, future studies are warranted to unravel whether targeting T lymphopenia can improve cardiac repair and function following MI. Since glucocorticoids mediate both T-cell trafficking to the bone marrow and thymocyte death, blocking glucocorticoid signaling would probably prevent T-cell defects. However, glucocorticoids are important to a plethora of physiological processes, including metabolism, immune response, cardiovascular function, skeletal growth, and reproduction (50). Blocking systemic glucocorticoids may have serious side effects. One viable approach will be to generate a mouse line with thymocyte- and T cell-specific deletion of glucocorticoid receptor and determine whether this would ameliorate post-MI cardiac repair and function by preventing MI-induced lymphopenia and T-lymphopoiesis impairment.

In infection- or injury-induced immunosuppression, T cells not only show marked loss but also display functional impairment. Postseptic T cells exhibit a phenotype of anergy and exhaustion, evidenced by a decrease in cytokine production and an upregulation of inhibitory cell surface proteins (51). T cells from severe COVID-19 demonstrate impaired proliferation and enhanced susceptibility to activation-triggered cell death after antigen exposure (52). After stimulation with concanavalin A, T cells from mice at 21 days after infection with *Angiostrongylus cantonensis* exhibit a significant decrease in proliferation and IFN-γ generation (17). In cardiac arrest and resuscitation, peripheral T cells also demonstrate fewer IFN-γ and TNF-α after stimulation (16). Interestingly, our study showed that splenic T cells from *day 7* post-MI mice did not show impairment in the generation of cytokines after stimulation with PMA and ionomycin. Future studies are warranted to determine whether MI affects other functions of T cells, including proliferation and chemotaxis.

There are a few limitations in our study. First, MI primarily occurs in older population (53), and the thymus undergoes dramatic involution with aging (10). This study, however, used middle-aged mice (4–8 mo old). Whether aged mice show thymic dysfunction following MI needs to be investigated in future studies. Second, in a clinical setting, most patients with MI receive reperfusion therapy via percutaneous coronary intervention, pharmacological thrombolysis, or coronary artery bypass grafting (53). However, our study used a nonreperfused MI model that does not mimic most patients. Therefore, future studies are warranted to validate the findings revealed in this study using the myocardial ischemia/reperfusion model. Third, patients with MI have a higher chance of infection (54). Although our previous and current studies show both

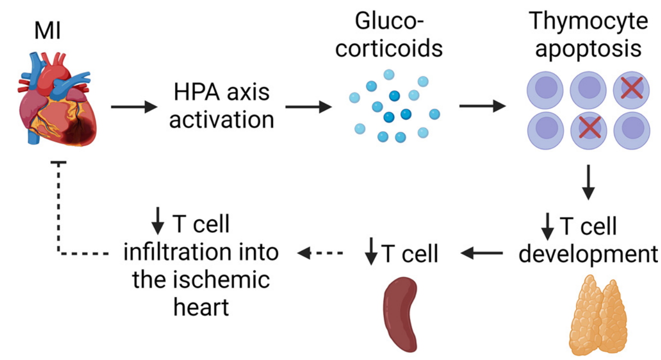

Figure 8. Diagram showing how myocardial infarction (MI) impairs T lymphopoiesis. MI activates the hypothalamic-pituitary-adrenal (HPA) axis and induces glucocorticoid secretion. Elevated glucocorticoids subsequently trigger thymocyte apoptosis and impair T development, causing smaller T-cell reservoir in the spleen. Decreased T-cell pool in the spleen may limit sufficient T cells infiltrating the ischemic heart, thus impeding cardiac wound healing. The dashed line indicates where more evidence is needed. Images were created with a licensed version of BioRender.com.

T lymphopenia and T lymphopoiesis impairment after MI (26), whether this can increase the risk and mortality of secondary infection remains unknown. Fourth, although our data show that glucocorticoid supplementation to ADX + MI mice reinduces thymic injury and dysfunction, whether other hormones secreted by adrenal glands play a role needs to be evaluated.

In summary, our study shows that MI induces thymic injury and dysfunction, causing smaller T-cell reservoirs in the spleen. The MI-induced thymic injury mainly results from the apoptosis of developing thymocytes, which is caused by the HPA axis activation and increased glucocorticoid secretion (Fig. 8). Reduced T-cell storage in the spleen may restrict sufficient T cells migrating to the ischemic heart, thus impairing cardiac wound healing.

DATA AVAILABILITY

Data will be made available upon reasonable request.

SUPPLEMENTAL MATERIAL

Supplemental Table S1 and Supplemental Figs. S1–S2: https://doi.org/10.6084/m9.figshare.25499596.v4.

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DISCLAIMERS

The content is solely the authors' responsibility and does not necessarily represent the official views of the University of South Florida.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

Y.M. conceived and designed research; D.X.S., D.M.K., H.R., and Y.M. performed experiments; D.X.S. and Y.M. analyzed data; D.X.S., D.M.K., H.R., S.Y.Y., and Y.M. interpreted results of experiments; Y.M. prepared figures; D.X.S. and Y.M. drafted manuscript; D.X.S., D.M.K., H.R., S.Y.Y., and Y.M. edited and revised manuscript; D.X.S., D.M.K., H.R., S.Y.Y., and Y.M. approved final version of manuscript.

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